

QSel: Budget-Aware Quality Selection for Dense Apple Phenology Detection

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Abstract. The DeepPhenoTree–Apple Edition Data Challenge requires detection of small, overlapping flowers, fruitlets, and fruit under a 30-minute inference budget. We present the system that placed third in its blind phase. A YOLO-World detector generates candidate boxes, and image-level classifiers determine the phenological stage. The detector often located objects approximately, but its scores did not reliably identify the most precisely localized boxes. We therefore introduce QSel, a lightweight post-processing model that estimates localization quality and ranks detections under the output limit. On a fixed out-of-fold candidate set, QSel increased $\text{mAP}_{50:95}$ from 0.5192 to 0.5265 at a cap of 400. The complete system achieved 0.4962 on the hidden test set, ranking third among 12 visible blind-phase entries as of 2026-07-07. Quality-aware ranking thus provided a modest improvement under a fixed prediction budget.

Keywords: Plant phenotyping · Object detection · Apple orchards · Quality-aware rescoring

1 Introduction and Challenge Setting

Monitoring when apple trees flower and develop fruit is important for phenotyping, yield estimation, and orchard management. Automating this process requires detecting hundreds of small, visually similar objects in high-resolution images. The challenge is not only to find each object, but also to place a tight bounding box around it while operating under limited inference resources.

We study this setting through the DeepPhenoTree–Apple Edition Data Challenge, associated with CVPPA 2026. The underlying RGB dataset was collected across European orchards [4]. Participants submit one detector for three phenological stages: flowering, fruitlet, and fruit. The released challenge split contains 558 labeled public images, the online-development phase has 86 images, and the final blind test contains 164 images. Each challenge image is effectively single-stage.

The official metric is COCO-style mean average precision, averaged over intersection-over-union (IoU) thresholds from 0.50 to 0.95 [3]. A loosely placed box may therefore count at the lowest threshold but fail at stricter thresholds. The runtime budget is 30 minutes on one NVIDIA V100 GPU, and the evaluation

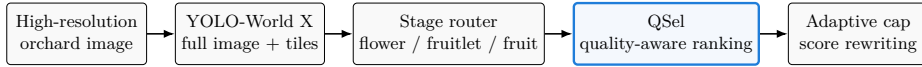


Fig. 1: Runtime system. YOLO-World X produces full-image and tiled candidates; a classifier cascade maps the image to one stage; QSel ranks candidates by predicted localization quality; and an adaptive cap produces the final detections.

package limits the number of submitted detections per image. These constraints make localization quality and candidate ordering important in addition to object discovery.

Our development trajectory began with detector scaling, longer training, label cleaning, and stage-specific experiments. These improved the baseline but left a persistent gap between AP_{50} and $mAP_{50:95}$. On local validation, before cap selection, at least one routed candidate overlapped 94.6% of ground-truth objects at IoU 0.50, but only 62.3% at IoU 0.75. After selecting 400 predictions, IoU-0.75 coverage remained 61.3%. Thus, deletion by the cap explained little of the high-IoU gap: many objects were found approximately, but the available boxes were not localized precisely enough. Raw detector confidence also did not reliably order the best-localized candidates, a problem related to prior quality-aware detection objectives [2].

This paper is primarily a challenge-solution report, not a new end-to-end detection objective. Its contributions are: (1) a third-place system for budget-constrained apple phenology detection; (2) a diagnosis of localization and ordering bottlenecks; and (3) a controlled comparison in which post-hoc QSel raised mean AP from 0.5192 to 0.5265 on an identical candidate pool and cap.

The data also required cleaning. We trained from a 497-image clean base with 86,115 boxes, excluding 61 images with rare but severe wrong or globally offset annotations. We then performed conservative coordinate-only consensus relabeling that snapped boxes to teacher consensus without adding boxes, deleting boxes, or changing stages.

2 Method

Figure 1 summarizes the submitted system. A detector generates full-image and tiled candidates. An image-level router selects the phenological stage, after which QSel ranks the compatible candidates and rewrites their scores before the final output cap.

2.1 Detector and Stage Routing

The detector is YOLO-World X fine-tuned for the three stages [1]; its pre-training supplied a broad visual prior while fitting the V100 budget. Training used consensus-reabeled 1536-pixel tiles with 0.35 overlap, YOLOv8x-worldv2 initialization, SGD, batch size 3 (nominal 128), loss weights 10.0/0.5/1.5 for box/class/distribution terms, mosaic and automatic mixed precision disabled,

and hard-box weighting over IoU 0.50–0.75. We submitted one checkpoint from a later hard-tail continuation.

At inference, the wrapper used a 1792-pixel full-image pass plus 1536-pixel tiles with 0.30 overlap. The confidence threshold was 0.05; detector and class-agnostic merge non-maximum suppression used IoU 0.60 and 0.50, respectively.

Image-level routing is performed after candidate generation because challenge images are effectively single-stage. A binary flower/non-flower ResNet-18 is evaluated first. If the image is non-flower, an eight-model fruit/fruitlet ensemble combines five ResNet-18 folds with ConvNeXt-Tiny, EfficientNetV2-S, and Swin-T classifiers. On out-of-fold public records, the flower/non-flower classifier was correct on 543/543 available examples, and the fruit/fruitlet ensemble was correct on 267/271. Routing filters detector classes incompatible with the predicted image stage and emits retained boxes as that stage.

2.2 QSel: Post-hoc Quality Selection

QSel turns dense inference into budgeted ranking. For candidate i with box b_i , detector score s_i , and tabular features x_i , the target is the average same-class matching success across the ten COCO IoU thresholds:

$$q_i = \frac{1}{10} \sum_{t \in \{0.50, 0.55, \dots, 0.95\}} \mathbf{1}[\text{IoU}(b_i, \text{GT}) \geq t]. \quad (1)$$

Matching is greedy and one-to-one, so a duplicate remaining after a higher-ranked match receives no positive target.

QSel uses compact histogram-gradient-boosted tree models trained on five selector-level image folds over local candidate records. Candidate features include detector score and rank, box geometry, normalized position, edge distance, image statistics, class and resolution indicators, class share, and duplicate-overlap context. The submitted package contains a legacy rank selector and a 320-tree soft-quality regressor. The regressor predicts q_i^{sel} , an estimate of q_i , which is used for cap ranking, tail release, and score rewriting.

The submitted score rewrite was a geometric blend:

$$s_i^{\text{final}} = (s_i^{\text{det}})^{0.60} (q_i^{\text{sel}})^{0.40}. \quad (2)$$

Both inputs lie in $[0, 1]$. The detector term retains evidence that the candidate represents an object; the selector term favors tighter localization and reduces the priority of duplicates. The 0.60/0.40 exponents were selected during local out-of-fold development, not from a theoretical criterion. The hidden submission used a base cap of 392 detections, followed by guarded stage- and resolution-dependent releases and a soft-QSel tail gate for dense images. After selection, scores were rewritten with the same blend.

A complete weight-sensitivity curve was not retained, and the records do not isolate ranking from score rewriting. We therefore make neither a transferability claim for 0.60/0.40 nor a general calibration claim.

3 Results

The complete hidden-test system scored 0.4962 $\text{mAP}_{50:95}$ and finished third among 12 visible blind-phase leaderboard entries as of 2026-07-07. Archived run-time measurements extrapolated to 23.5 minutes on the 164-image blind set, below the 30-minute V100 budget.

Table 1 visibly separates official milestones from the controlled local comparison. Scores from different splits are not directly comparable and must not be read as ablations of QSel.

Table 1: Results. Panel A documents historical official milestones that used different systems and splits. Panel B isolates ranking and score rewriting on one fixed out-of-fold candidate pool with cap 400; Δ is relative to raw detector confidence.

A. Official challenge milestones				
System milestone	Split	Cap	mAP _{50:95}	
Plain detector, weak count control	Online development	–	0.4744	
Cap 400 with hand reranking	Online development	400	0.4833	
QSel-based pipeline	Online development	400	0.4921	
Complete submitted system	Hidden test	392 + tail	0.4962	
B. Controlled local comparison				
Ranking and score method	AP ₅₀	AP ₇₅	mAP _{50:95}	Δ
Raw detector confidence	0.8892	0.5395	0.5192	–
QSel rank + score rewrite	0.8927	0.5494	0.5265	+0.0073
Best recorded soft-quality sweep	0.8936	0.5521	0.5291	+0.0099

On the fixed local pool, QSel improved $\text{mAP}_{50:95}$ by 0.0073 and AP_{75} by 0.0099 over raw confidence, with a positive $\text{mAP}_{50:95}$ difference in all five selector folds. The best recorded sweep was not the exact hidden configuration. These are modest gains from one fold partition; no multi-seed interval was established. The hidden score evaluates the entire pipeline and therefore does not isolate QSel.

4 Discussion

Local diagnostics identify small-object high-IoU localization as the remaining bottleneck. On local validation, 6,437 of 12,265 boxes (52.5%) have square-root area below 48 pixels. Flowering was hardest: local $\text{mAP}_{50:95}$ was 0.4613 for flowering, 0.5308 for fruitlet, and 0.6448 for fruit. Simple confidence thresholds were unsafe because archived online-development predictions retained 4,211 rows below confidence 0.10 after rewriting, including true positives. QSel therefore acts

as a ranking and score-rewriting mechanism rather than a deletion rule or a claim of probabilistic calibration.

QSel operates only after inference and does not modify the detector’s losses. Future comparisons should train a detector with quality-aware targets [2] and add rank-only, rewrite-only, weight-sensitivity, repeated-seed, and cross-orchard evaluations. The practical lesson is narrow: under a strict cap, post-hoc selection can allocate output slots more effectively, but it complements rather than replaces better localization.

5 Conclusion

We presented the third-place DeepPhenoTree–Apple Edition system for dense phenology detection under a strict inference budget. The detector often discovered the correct objects but struggled to rank candidates by precise localization quality. QSel rescored and selected frozen outputs, producing a modest controlled local improvement. The remaining gap at strict overlap thresholds shows that rescoring cannot replace better localization; future work should compare post-hoc selection with jointly trained quality-aware detectors across orchards and imaging conditions.

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